

# Static Cosmology

## Lecture 7: Medium Density and Unluminous Mass

Haeon Jeong

2026-10-12–2026-10-18

---

### Overview

Rotation curves are inverted for the sum of luminous and medium density under a Jeans-equilibrium condition.

### 1 Observational setup

Source coordinates, observing band, selection function, and calibration state are fixed at the outset.

$$v_c^2/r = d\Phi_{eff}/dr$$

### 2 Transport or equilibrium equation

The governing equation is solved first in a homogeneous limit and then with the stated spatial dependence.

$$\nabla^2\Phi_{eff} = 4\pi G(\rho_{lum} + \rho_m)$$

### 3 Connection to data

Each model quantity is tied to a measured flux, spectrum, angle, count, or temperature.

$$\nabla(\rho_m\sigma_m^2) + \rho_m\nabla\Phi = 0$$

### 4 Degeneracies

Calibration, population, and medium parameters are varied separately to expose their degeneracies.

### 5 Example

The worked example carries units and uncertainty from the input catalog to the reported result.

## Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

## Reading

- Jeong, ch. 7
- CMT density block
- ETHERMAP inversion note